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# OSL PRACTICAL 1
# Basic File and Directory Operations in Linux

# 1. Display current working directory
pwd

# Display all files, including hidden files
ls -la

# 2. Create main directory and three subdirectories
cd ~/OSL_Practicals
mkdir -p OS_Lab
cd OS_Lab

mkdir -p Programs Documents Backup

# Display directories
ls
# 3. Create text files
touch file1.txt file2.txt file3.txt

# Write content into files
echo "This is the first file." > file1.txt
echo "This is the second file." > file2.txt
echo "This is the third file." > file3.txt

# Display file contents
cat file1.txt
cat file2.txt
cat file3.txt

# 4. Copy a file
cp file1.txt file1_copy.txt

# Rename a file
mv file2.txt renamed_file.txt

# Move a file into Documents
mv file3.txt Documents/

# Delete a copied file
rm file1_copy.txt

# Display remaining files
ls

# Display Documents directory contents
ls Documents

# 5. Search for a file
find . -name "renamed_file.txt"
# Display detailed file information
stat renamed_file.txt
# Display file size
du -h renamed_file.txt

# Display complete directory structure
ls -lR
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# OSL PRACTICAL 2
# File Permissions, Ownership, Process Monitoring and System Resources

# 1. Create folder and files
cd ~/OSL_Practicals
mkdir -p Practical2
cd Practical2

touch file1.txt file2.txt file3.txt

# 2. Set numeric permissions
chmod 644 file1.txt
chmod 755 file2.txt
chmod 700 file3.txt
ls -l

# 3. Modify permissions using symbolic mode
chmod u+x file1.txt
chmod g-w file2.txt
chmod a+r file3.txt
ls -l

# 4. Identify current user and group
whoami
id
ls -l

# 5. Change file ownership
sudo chown root:root file1.txt
ls -l file1.txt

# Restore ownership
sudo chown aryan:aryan file1.txt
ls -l file1.txt

# 6. Display running processes
ps
ps aux
pgrep -a bash
echo $$

# 7. Monitor processes in real time
top
# Press q to exit top

# 8. Display CPU information
lscpu

# 9. Display memory usage
free -h

# 10. Display disk-space information
df -h

# 11. Display Practical 2 folder size
du -sh ~/OSL_Practicals/Practical2

# OSL PRACTICAL 3
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# Student Records Management Using Linux Commands

cd ~/OSL_Practicals

# 1. Create directories
mkdir -p Student_Records
cd Student_Records
mkdir -p FY SY TE BE Backup

# 2. Create student record files
echo -e "Name: Rahul\nRoll No: 101\nName: Priya\nRoll No: 102" >
FY/FY_Students.txt

echo -e "Name: Amit\nRoll No: 201\nName: Sneha\nRoll No: 202" >
SY/SY_Students.txt

echo -e "Name: Karan\nRoll No: 301\nName: Neha\nRoll No: 302" >
TE/TE_Students.txt

echo -e "Name: Akash\nRoll No: 401\nName: Riya\nRoll No: 402" >
BE/BE_Students.txt

# Display student records
cat FY/FY_Students.txt
cat SY/SY_Students.txt
cat TE/TE_Students.txt
cat BE/BE_Students.txt

# 3. Copy selected files
cp FY/FY_Students.txt Backup/
cp SY/SY_Students.txt Backup/

# 4. Rename and delete files
mv Backup/FY_Students.txt Backup/FY_Students_Copy.txt

touch TE/Temporary.txt
rm TE/Temporary.txt

# 5. Display complete directory structure
ls -R

# Optional tree view
tree

# OSL PRACTICAL 4
# File Permissions and Ownership in Linux

# 1. Create project directory and files
cd ~/OSL_Practicals
mkdir -p AIML_Project
cd AIML_Project
touch students.txt faculty.txt administrators.txt
# Add content
echo "Student records: Rahul, Priya, Amit" > students.txt
echo "Faculty records: Prof. Sharma, Prof. Patil" > faculty.txt
echo "Administrator records: Admin1, Admin2" > administrators.txt

# 2. Assign different numeric permissions
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chmod 644 students.txt
chmod 640 faculty.txt
chmod 600 administrators.txt

# 3. Display ownership and permissions
whoami
id
ls -l
stat students.txt
stat faculty.txt
stat administrators.txt
# 4. Change permissions using symbolic notation
chmod u+x students.txt
chmod g+w faculty.txt
chmod o+r administrators.txt
ls -l

# Change permissions using numeric notation
chmod 755 students.txt
chmod 644 faculty.txt
chmod 700 administrators.txt
ls -l

# 5A. Demonstrate read permission
cat students.txt
chmod 000 students.txt
cat students.txt
chmod 644 students.txt

# 5B. Demonstrate write permission
echo "New student: Karan" >> students.txt
cat students.txt
chmod 444 students.txt
echo "New student: Neha" >> students.txt
chmod 644 students.txt

# 5C. Demonstrate execute permission
echo 'echo "AIML Project Executed Successfully"' > test.sh
chmod 644 test.sh
./test.sh
chmod 755 test.sh
./test.sh
# Final permission details
ls -l

# OSL PRACTICAL 5
# Shell Scripting

cd ~/OSL_Practicals
mkdir -p Practical5
cd Practical5

# 1. Multiplication table using FOR, WHILE and UNTIL

cat > multiplication.sh << 'EOF'
#!/bin/bash

echo "Enter a number:"
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read num

echo "Using FOR loop:"
for ((i=1; i<=10; i++))
do
    echo "$num x $i = $((num*i))"
done

echo "Using WHILE loop:"
i=1
while [ $i -le 10 ]
do
    echo "$num x $i = $((num*i))"
    ((i++))
done

echo "Using UNTIL loop:"
i=1
until [ $i -gt 10 ]
do
    echo "$num x $i = $((num*i))"
    ((i++))
done
EOF

# 2. Positive, negative or zero

cat > number_check.sh << 'EOF'
#!/bin/bash

echo "Enter a number:"
read num

if [ "$num" -gt 0 ]
then
    echo "The number is POSITIVE."
elif [ "$num" -lt 0 ]
then
    echo "The number is NEGATIVE."
else
    echo "The number is ZERO."
fi
EOF

# 3. File existence and information

cat > file_info.sh << 'EOF'
#!/bin/bash

echo "Enter file name:"
read filename

if [ -e "$filename" ]
then
    echo "File exists."
    echo "File information:"
    ls -l "$filename"
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    echo "File size:"
    du -h "$filename"

    echo "Detailed information:"
    stat "$filename"
else
    echo "File does not exist."
fi
EOF

# Give execute permission
chmod +x multiplication.sh number_check.sh file_info.sh

# Execute scripts one by one
./multiplication.sh
./number_check.sh
./file_info.sh

# Display all files
ls -l
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